

DATA SCIENCE RETREAT

PRODUCT DEMO MVP - OCT 2025

MADE BY GRANT THOMAS

Mentored by Krzysztof Buzar

AI MUSIC MENTOR

UNBLOCK YOUR CREATIVE PROCESS

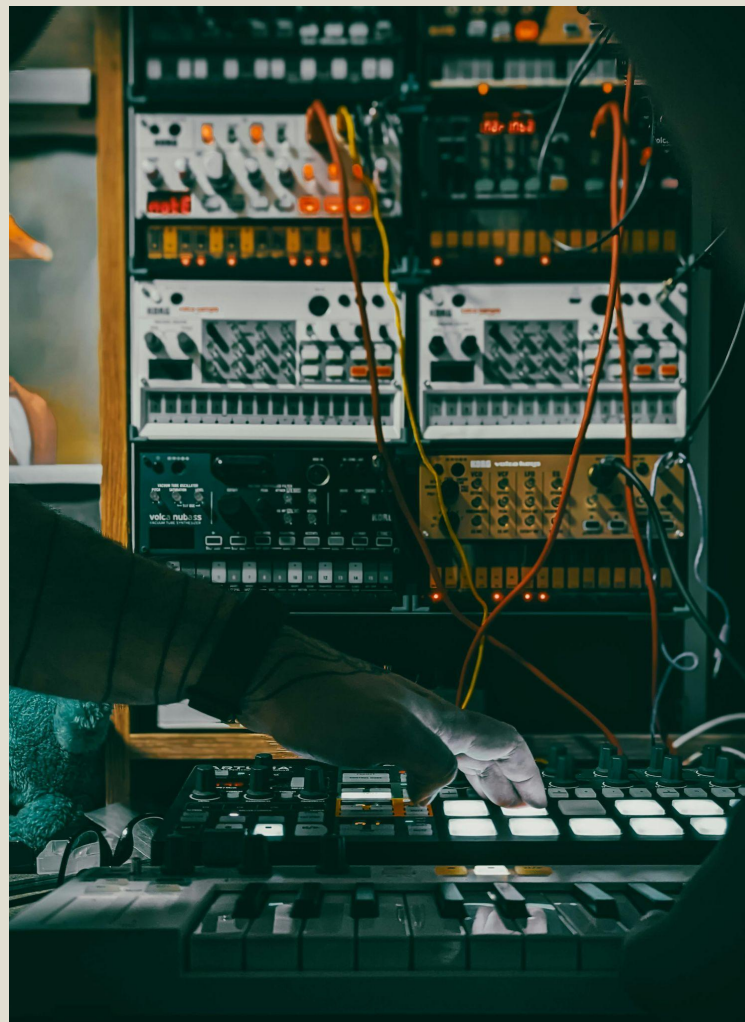


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WHO

Grant AKA 16 Faces

Software Engineer
Berlin Based Producer
DIY Label Owner
Runs Community Music Feedback Workshops

WHO AM I?



Beginner Producers need help finishing their songs

The Status Quo in tooling is

- Generic (Youtube/generic example tools)
- Takes away musical agency (Generative Audio AI)

And real credible human help is

- Hard to Access
- Expensive



CORE PROBLEM

Writer's Block is consistently ranked in top 3 problems for young musicians across survey

20-60% of Musicians already use AI in some way during their process despite skepticism - [ArtSmart.AI](#)

82% Musician believe Human creativity to essential to music making - BPI Survey

250+ threads on reddit producers asking for help on their music and many more across Discord

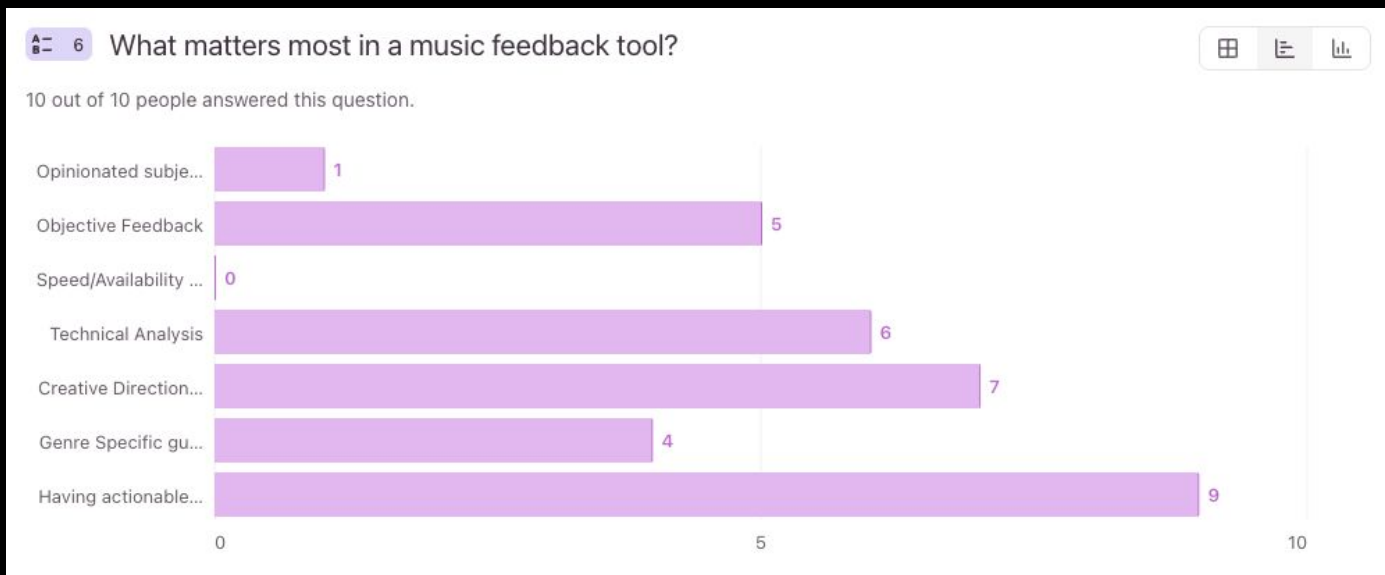
"Sometimes it's nice to borrow someone's ear and that isn't always accessible. What might be nice is if it offered references to existing songs that might be relevant,"

"Really interesting use of ai, I like how it is up to you to action the changes"

"I think it has to be genuine, can't sound like chat-gpt giving you the answers, and probably needs to be specific to a genre in terms of the insights"

"Privacy would be an important thing to consider. Being sure the music is not stored anywhere, and that the sessions remain private."

MARKET SURVEY



CORE INSIGHT - HAVING ACTIONABLE
FEEDBACK IS KEY

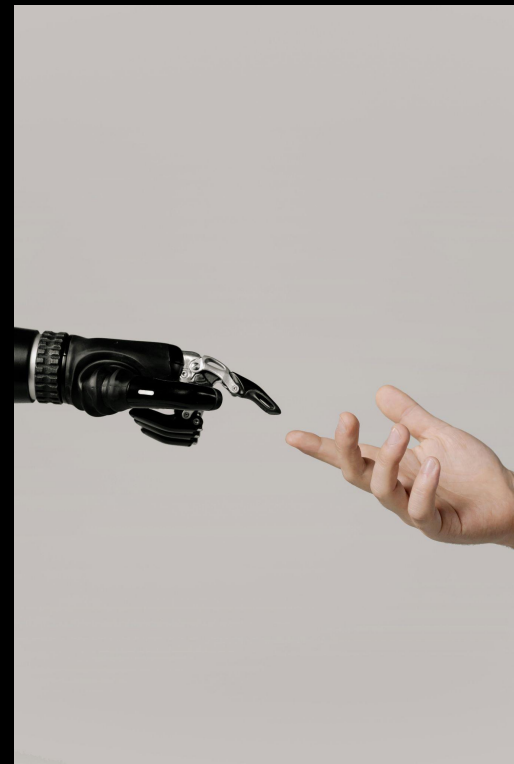
MARKET SURVEY

AI MUSIC MENTOR - TECHNO EDITION

Get Unstuck with Practical AI Feedback
In the tone and style of a local Berlin
Producer 16 Faces

- Simple practical Feedback
- Specific and Sound aware
- With Arrangement Visualizations
- Natural language
- Without complex technical jargon

SOLUTION



ARRANGEMENT

RHYTHM

Drums and percussion

O: INTRO/OUTRO

ENERGY

Volume Intensity

A: GROOVE SECTION

FREQUENCY

EQ (Low, Mid Hi)

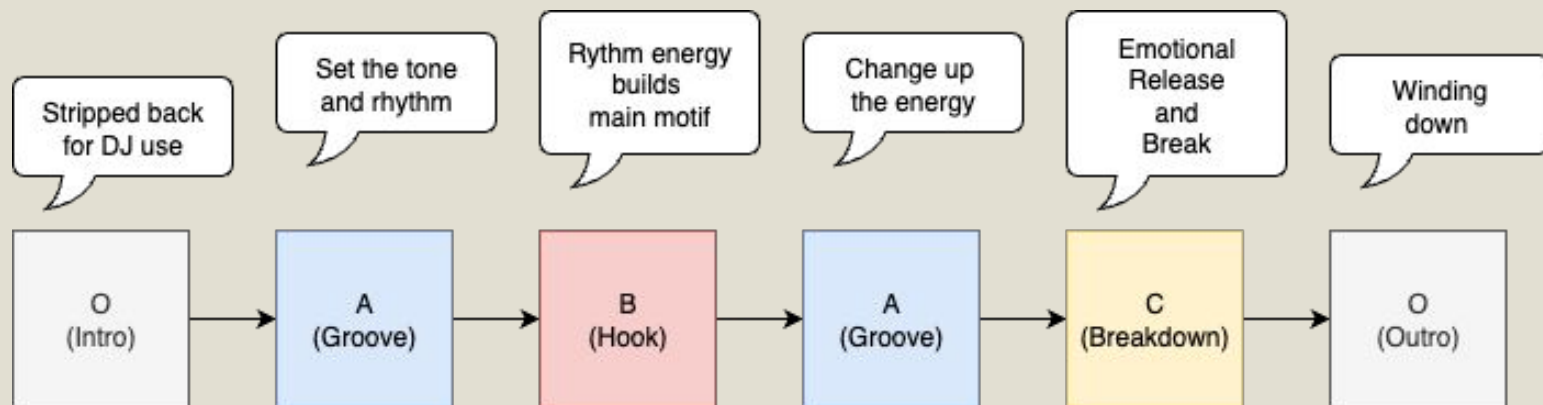
B: Main SECTION

C: BREAK DOWN

MUSIC THEORY

a quick lesson

RHYTHM AND ENERGY IS CORE TO TECHNO ARRANGEMENT



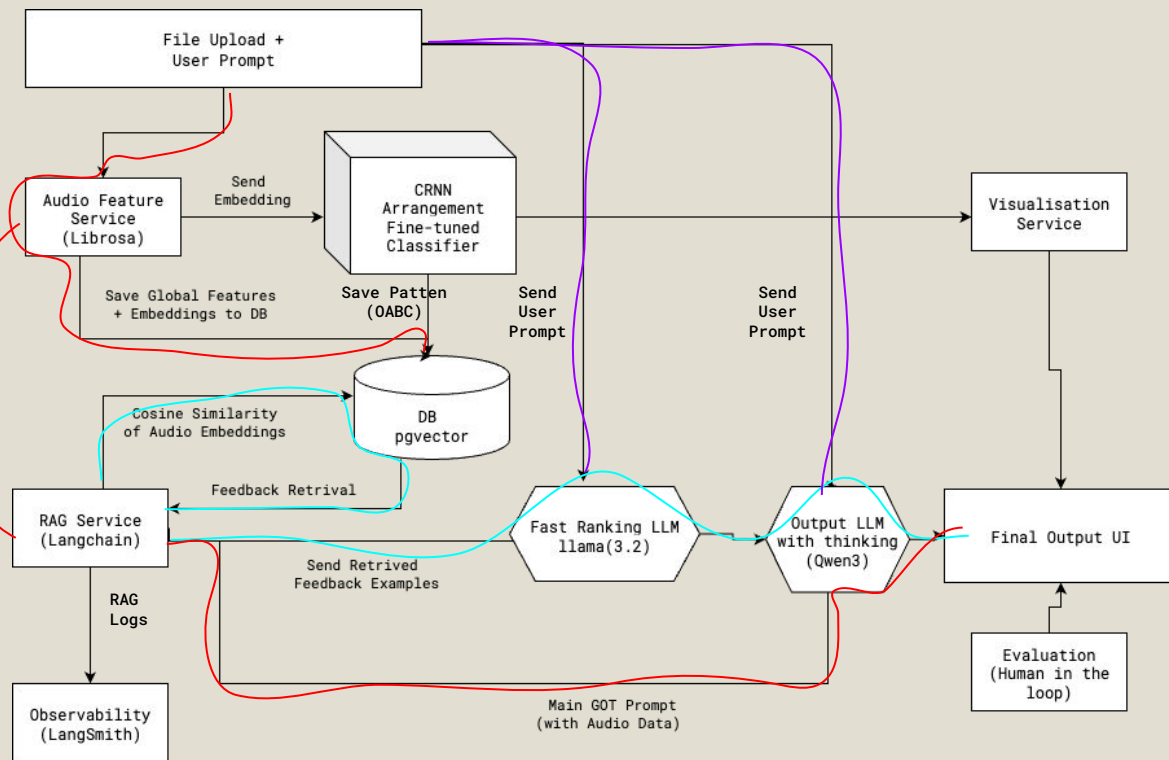
➔ **AN ARRANGEMENT EXAMPLE**

So I was using AI MUSIC MENTOR the other day and here's what it told me about my collab track

"there's room to carve out more space for clarity and dynamics. Add subtle percussion changes or a shifted groove will keep the energy alive."

Maybe AI MUSIC MENTOR can help us out again too!

DEMO TIME

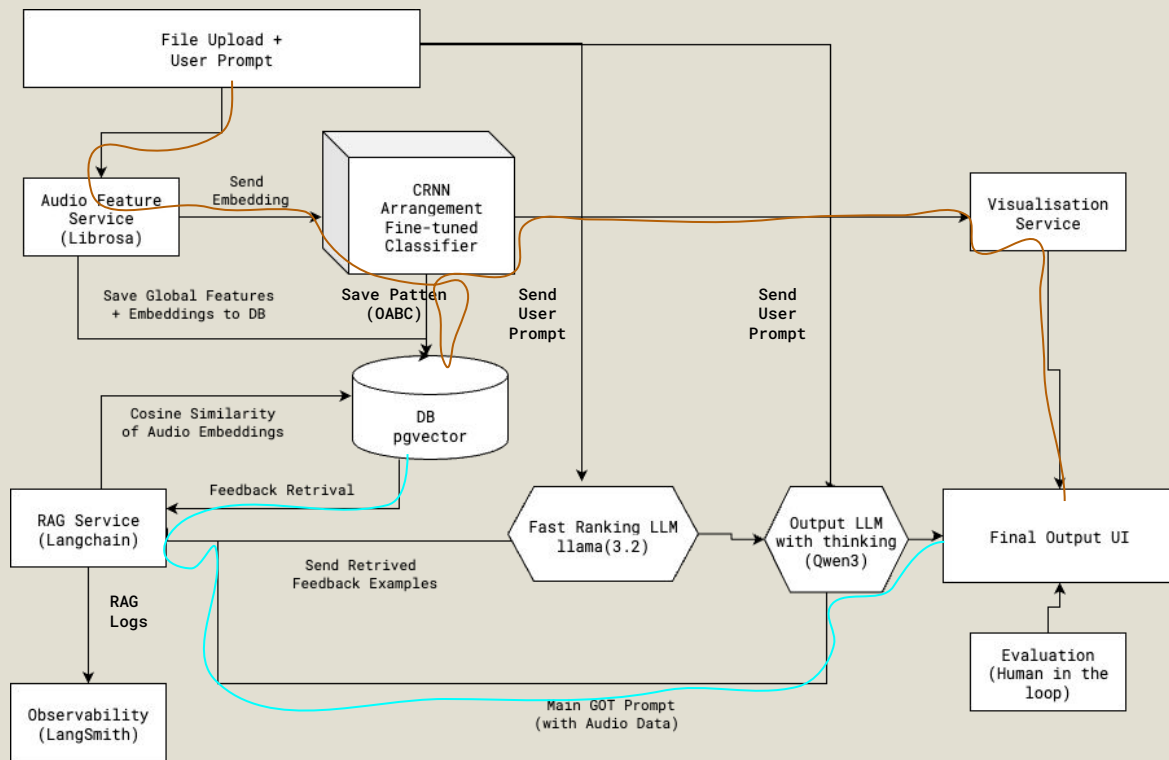


Part I

RAG System using Audio Embeddings

1. **Upload Audio**
(input + reference tracks)
2. **Extract Global Audio Features**
(19-dimensional)
3. **Embed Features**
(create embedding vector)
4. **Save Audio Data to DB**
(PostgreSQL + pgvector)
5. **Retrieve with Cosine Similar Songs**
(and feedback examples)
6. **Rank Retrieved Samples with Llama3.2**
(small LLM)
7. **Prompt Graph of Thought main LLM**
with context
8. **LLM Output from Qwen3:8b**
(final feedback)
9. **Human in the Loop Evaluation**

SYSTEM OVERVIEW PT I



Part II Arrangement Classification

- 1. Extract Audio Features**
(15-dimensional CRNN features)
- 2. Embed Features**
(combine RMS, mel, chroma, tempogram, MFCC)
- 3. Pad Embedding**
(standardize to 300 frames/meter, 201 meters max)
- 4. Send Features to CRNN**
(TensorFlow model)
- 5. Predict Classes in 5-second segments**
(O-A-B-C)
- 6. Post-Process Chunks**
into smooth segments
- 7. Visualize Segments on waveform**
with arrangement overlay

SYSTEM OVERVIEW PT II



Our Global features (using Librosa):

- Help describe musical aspects of the tracks for our RAG and LLM analysis
- We are also able to embed them into a single 19 Dimension vector for cosine similarity search

For our CRNN the features were kept the same
As the base model training, similar to our RAG
features

Rhythm, Harmony, Energy and Frequency to describe
Musicality per 5 second chunk for training
And inference.

Feature Extraction

```
"global_feature_data": {
  "metadata": {
    "duration": 58.181995464852605,
    "sample_rate": 22050
  },
  "rhythm": {
    "tempo": 132.51201923076923,
    "onset_density": 6.135918803535392,
    "syncopation_level": 0.2946557188130218,
    "rhythmic_variance": 0.013681105405780854,
    "beat_strength": 0.6000788807868958
  },
  "harmony": {
    ...
  },
  "energy": {
    ...
  },
  "spectral": {
    ...
  },
  "frequency": {
    ...
  }
}
```

DB example
of features
before we embed them

Fine-tuning a CRNN

with tensorflow

A CRNN is a Convolutional Recurrent Neural Network (Hybrid)

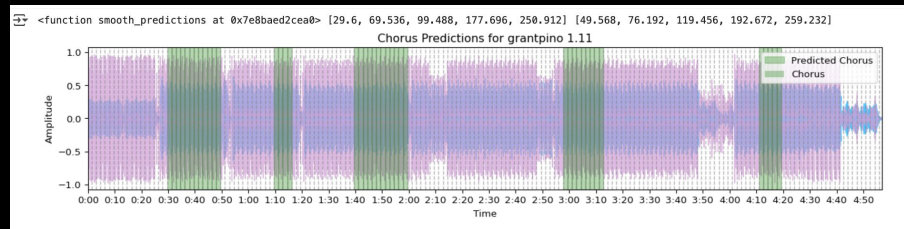
CNN (Convolutional Neural Network) - Good at extracting spatial features from data like images or spectrograms for audio

RNN (Recurrent Neural Network) - Good at modeling sequential/temporal patterns

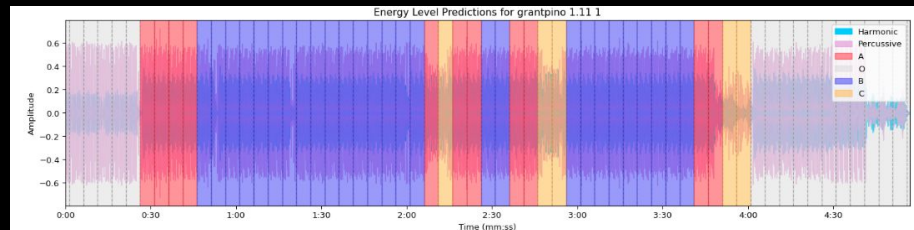
Our Model:

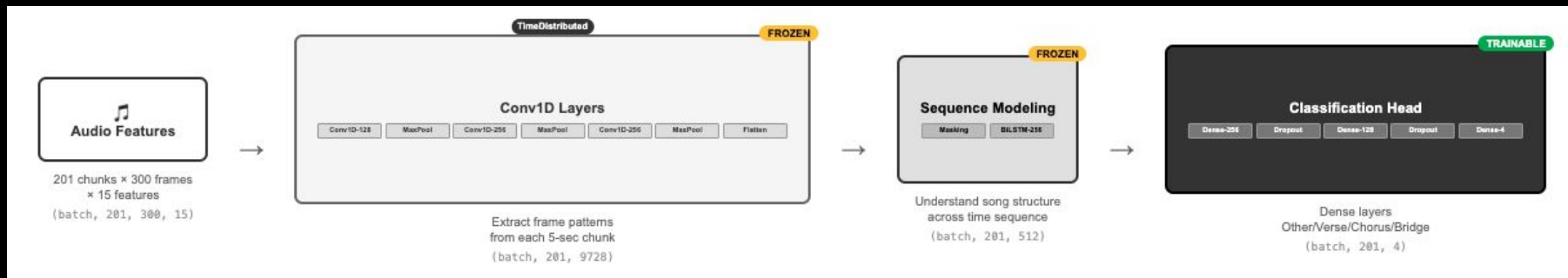
- Total Layers: 9 main layers + 8 CNN sub-layers within TimeDistributed
- Frozen Parameters: ~25M (entire CNN + LSTM backbone)
- Trainable Parameters: ~200K (new classification head only)
- Input Size: $201 \times 300 \times 15 = 904,500$ features per meter chunk
- Final Embedding: Each meter represented as 128-dimensional vector before classification (4 classes)

ORIGINAL CRNN MODEL



CRNN MODEL AFTER FINE TUNING





Librosa Embedding -> Extract patterns from chunks (5sec) -> understand chunk in context of a song -> predict probabilities of arrangement class per chunk

Embedding Transformations (with Batches) - 1 Song per batch - 5 seconds per chunk

Chunking the songs per 5 seconds gave the most reliable outcome to get the macro level segmentation without predicting very tiny micro sections

1. Meter Level: (batch, 201, 300, 15) → CNN → (batch, 201, 9728)
2. Song Level: (batch, 201, 9728) → LSTM → (batch, 201, 512)
3. Classification: (batch, 201, 512) → New Head Dense layers w dropout → (batch, 201, num_classes=4)
 - Final prediction for each 5-second chunk: 0 / A / B / C Classes

CRNN Architecture

TRAINING DATA

The base CRNN was trained on 332 songs with 2 classes (about 5 labels per song)

Approx 1660 class data points

For our fine-tuning I used 30 songs with 4 classes each with about 10 labels per song.

For fine-tuning I also augmented the dataset 4x per song by time-stretching the songs to different lengths to keep the harmonic content in tack.

Approx 1200 class data points

Since the class labelling was not even distributed I also weighted the classes during training to improve the accuracy metrics per class.

After running several tests on songs I knew well the 4 class version seemed more practical for advice even though 3 classes had better F1 scores in mosts of the classes.

Final Evaluation Metrics

**masked categorical cross-entropy
and masked accuracy**

4 Classes Fine-tuning

O(Other): Precision=0.707, Recall=0.629, **F1=0.665**

A(Main): Precision=0.441, Recall=0.519, **F1=0.477**

B(Breakdown): Precision=0.455, Recall=0.383, **F1=0.416**

C(Low): Precision=0.664, Recall=0.755, **F1=0.707**

3 Classes Fine-tuning

A (O & A): Precision=0.718, Recall=0.593, **F1=0.650**

B (Main): Precision=0.429, Recall=0.625, **F1=0.509**

C (Breakdown: Precision=0.893, Recall=0.622, **F1=0.733**

RAG STRUCTURE

LANG CHAIN for our RAG setup

LANG SMITH for observability

POSTGRES + pgvector DB

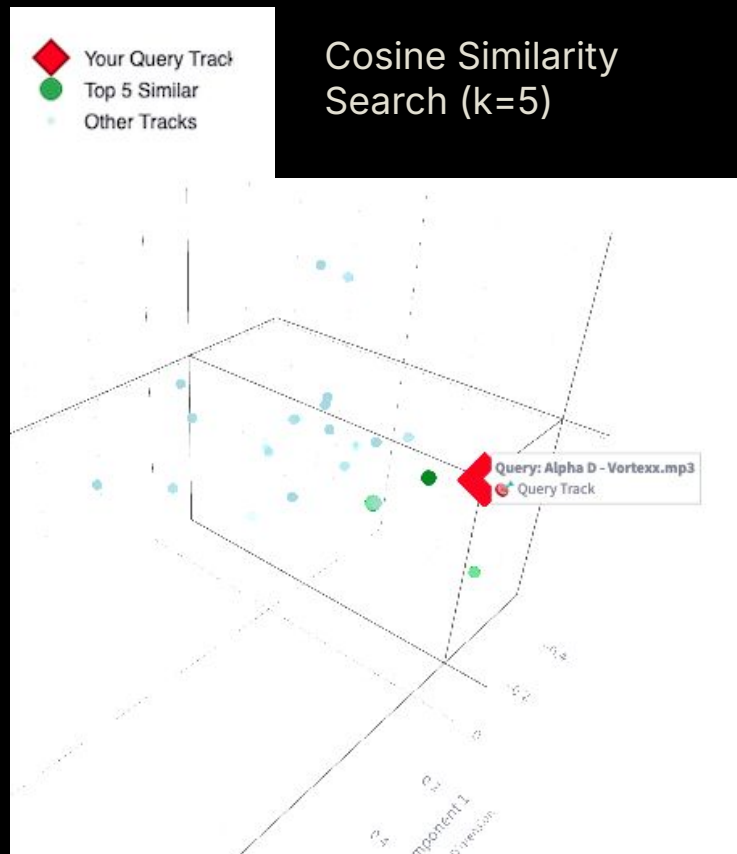
Search: **Cosine Similarity ON AUDIO EMBEDDINGS** -
(Euclidean didn't work well)
on Global Audio embeddings to find songs

Songs are connected to **Feedback entries**
which are then ranked by relevance to user's
prompt for filtering.

Ollama for LLM management

Llama 3.2 (3B) - **Ranking / Filtering Feedback**
decoder-only autoregressive transformer

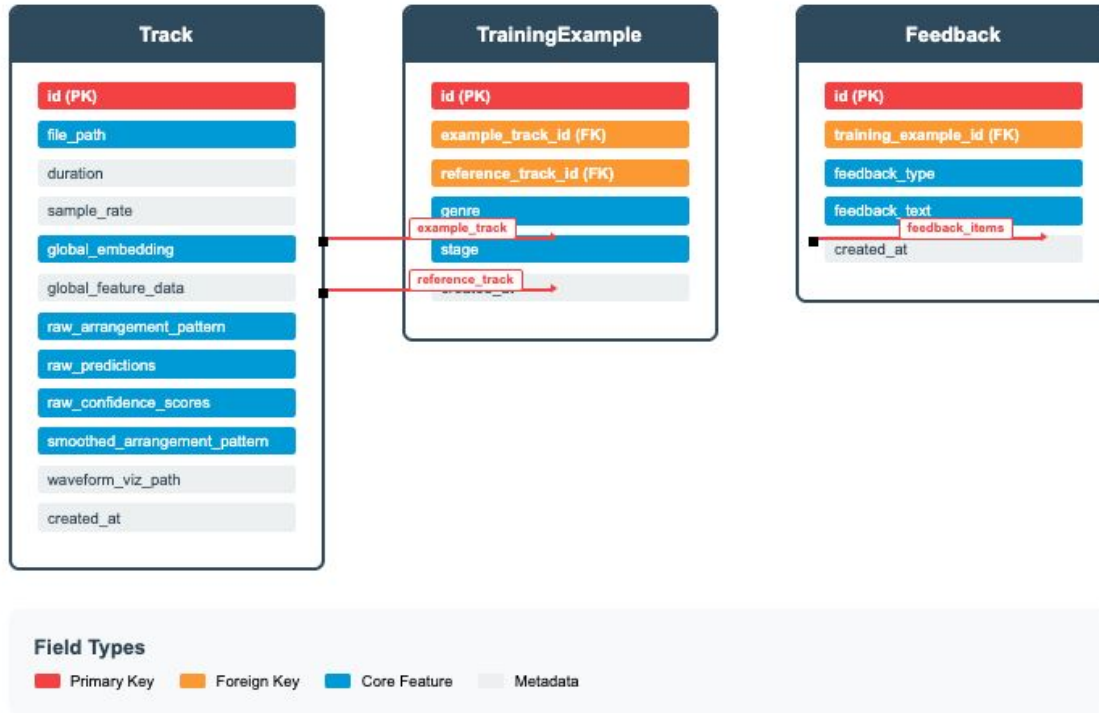
Qwen 3 with thinking (8B) - **Main Text Output**
(needed for Graph of Thought prompting)
decoder-only autoregressive transformer



Each Training Example
Has both an Example
& Reference Track

And is Attached to multiple
Feedback Entries

Hence when we retrieve a track
using cosine we can easily find
our associated feedback for
further filtering



RAG DB Schema

Our Main LLM Prompt used:

- Predicted Arrangement Pattern eg. (O-A-B-O)
- Global Audio Features (Librosa)
- Diff of features (input vs reference)
how far the input track is from ref track
- **Filtered Feedback examples from RAG (top 2)**
- Arrangement Section Definitions
- Our Producers musical rules and guidelines
- Main Goal for the output text
- + some guidance and rules

We also leverage **Graph of Thought** prompting for arrangement advice

Inspired by the AI Trackmate Paper

<https://arxiv.org/abs/2412.06617>

Primary goal: Help the user overcome writer's block by providing exactly 2 concrete, actionable suggestions that move their track forward. You are not trying to complete their track or provide a comprehensive roadmap - just give them the 2 most impactful, specific ideas to break through their current sticking point and continue creating. Keep feedback focused and digestible ...

- Final prompt snippet

Prompt Engineering

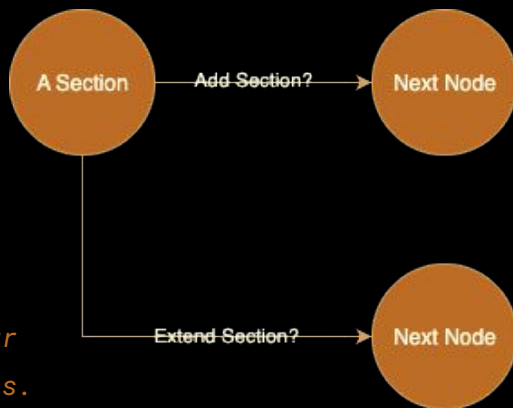
(much like herding cats)

In the prompt we Define Nodes
as arrangement sections eg (current step)

0, A, B, C

*This gives our system the
ability to reason one node at a time.*

*Eg. We have a section B that's very
short let's consider "Extend Section"
followed "add section" of node C as our
Reference track uses C after B sections.*



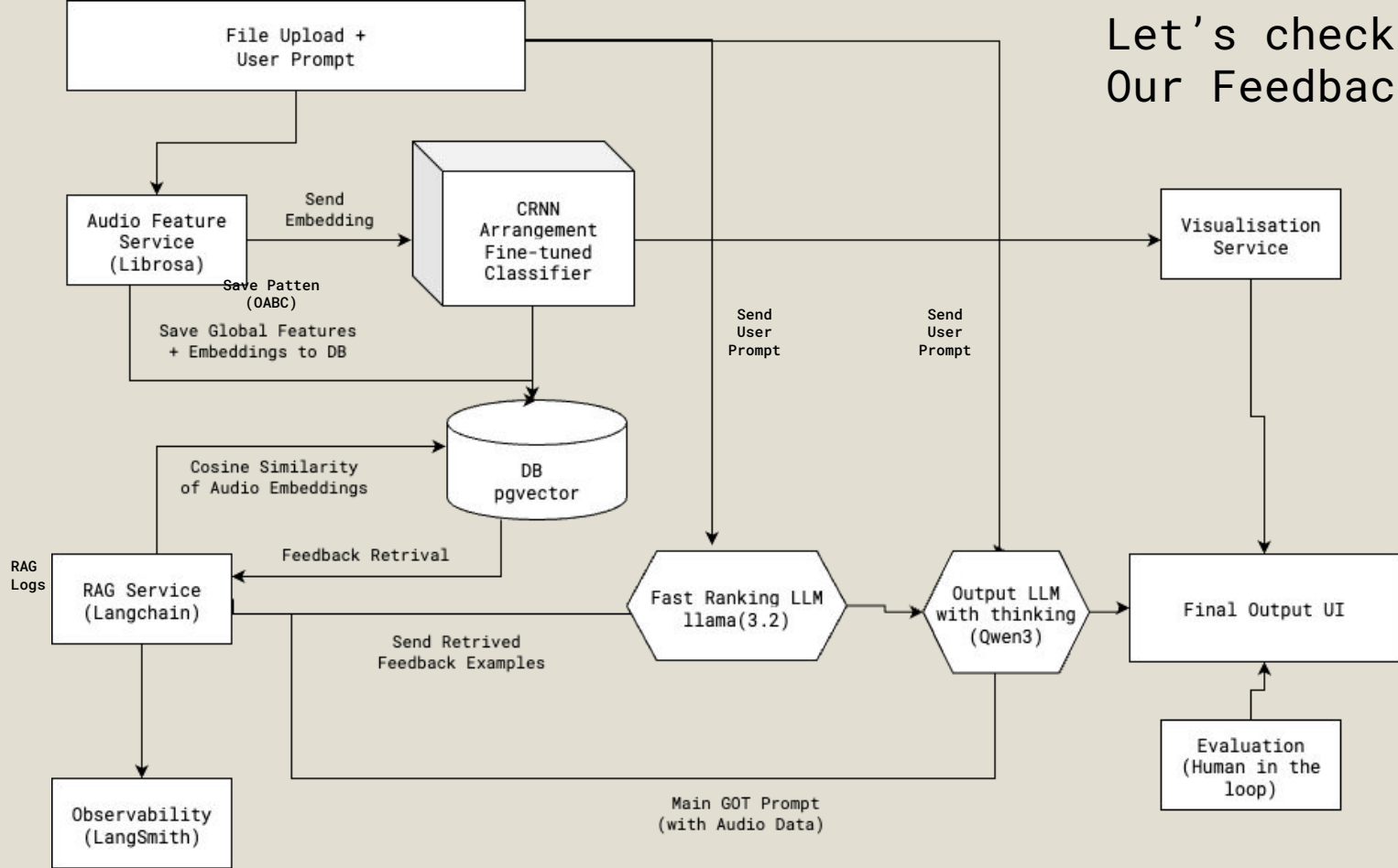
Edges are defined
as (decisions):

"add section",
"extend section",
"adjust energy",
"modify groove",
"mix advice",
"suggest classic track"

GRAPH OF THOUGHT

PROMPTING

Let's check on
Our Feedback!



2 Key Suggestions

1. Extend the B Section & Add a C Section for Dynamic Build-Up

Why it works:

The B section is your main hook, but it's currently underdeveloped compared to the reference track's B-A-B-A-C-O structure. By **extending the B section by 4 chunks (16 bars)**, you'll give it more room to breathe and build momentum. Then, **add a C section after the B** (like the reference track does) to introduce a new layer of rhythm or melody. This will create a clearer arc and keep the listener engaged.

Action Steps:

- Trim the B section by 2 chunks (8 bars) to make space for a new C section.
- Add a C section with a slightly higher energy groove (think: tighter percussion, brighter synths) to bridge the B and the final climax.

2. Boost Brightness & Carve Mid-Range Clarity

Why it works:

The track feels a bit muted in the high end, which is common in deep techno but can be balanced with some sparkle. The mid-range (around 200Hz) is also a bit heavy, which might be muddying the low-end presence.

Action Steps:

- **Add subtle high-end sparkle** (around 8-12kHz) to the synth leads and hats to lift the brightness without overpowering the groove.
- **Carve space around 200Hz** using a narrow Q filter to reduce muddiness while preserving the low-end punch.

Musical Analysis

Your current structure (O-C-O-A-B-O) is a great starting point, but it's missing the layered progression seen in the reference track. The **B section** feels like it's holding back energy, and the **C section** could be expanded to build tension before the climax. The track's **brightness** is a bit muted, and the **mid-range** feels crowded, which could use some EQ shaping to carve space.

FINAL THOUGHTS

You're on the right track with this deep techno vibe—just need to stretch the arrangement a bit and refine the mix. Take your time to experiment with the B and C sections, and don't forget to dial in that brightness and clarity. When you're ready, come back with a new version, and I'll be here to help you polish it to perfection. Keep that groove alive! 🎵

OUR RAG OUTPUT



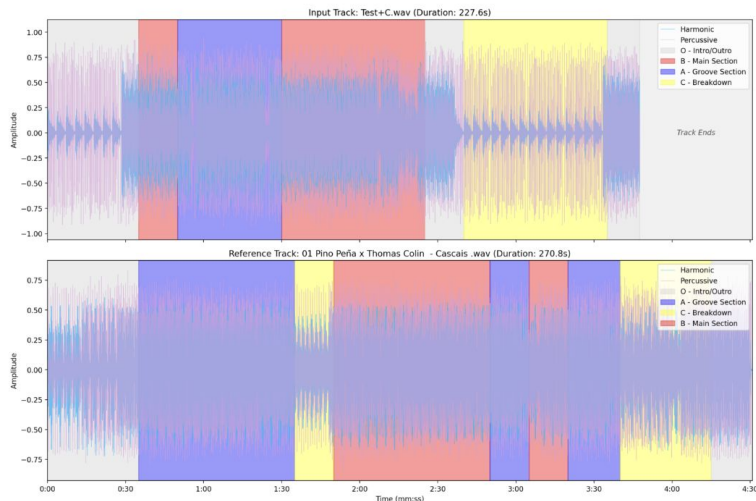
Here is what we made today

OUR REMIX

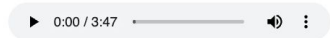
Arrangement Analysis

Input: 0-B-A-B-0-C-0 (135 BPM)

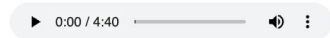
Reference: 0-A-C-B-A-B-A-C-0 (142 BPM)



Input Track Audio:



Reference Track Audio:



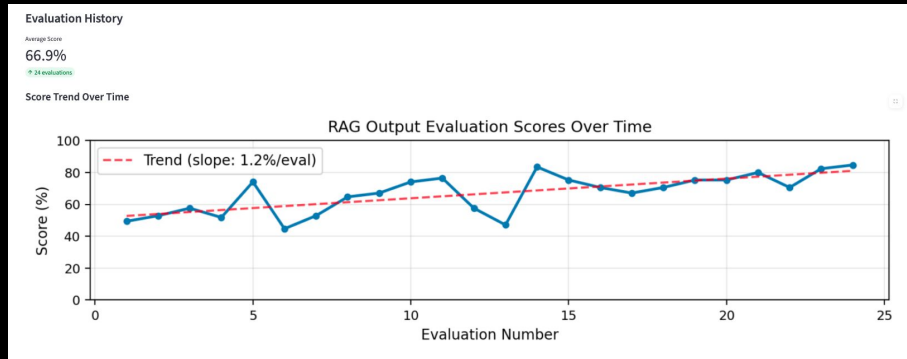
We add a C section
Which is detected

Our CRNN is working well

USING AI ADVICE

Weighted total score of output
as improved RAG (1 to 5)

- Audio Description Accuracy (Weight: 2x)
 - Practical Advice Quality (Weight: 2x)
 - Relevance to Input Audio (Weight: 1.5x)
 - Truthfulness (No Hallucinations) (Weight: 2x)
 - Technical Understanding (Weight: 1x)
- + Blind test with friends who produce music semi-professionally



EVALUATION

HUMAN OUTPUT (% ??)

Didn't have time to test
...

WHICH ADVICE WAS BETTER?

Survey asking 2 semi pro producers for blind
evaluation

Chat GPT 5
1.8 Trillion Parameters
29/50 ($\bar{x}$) = 58%

VS

AI MUSIC MENTOR
11 Billion Parameters
37/50 ($\bar{x}$) = 68%



CHALLENGES

Fine-tuning had limitations

Needed more data & time

Prompt Engineer is art not a science

Complexity of the problem
(features)

Song segmentation needed in DB
for meaningful comparison of song
chunks.

Filtering Feedback was not
straightforward.

Getting the producer tone into the
output

FUTURE PLANS

Extend Data with more diverse
songs for Fine-tuning

Mood and instrument classification
(Essentia C++)

Agentic workflow for better output

Deployment

Understand how to herd LLMs
(much like cats)

GET THE SYSTEM MORE ROBUST

INVOLVE AN ESTABLISHED TOP TIER PRODUCER FOR FEEDBACK TRAINING

INTEGRATING IT INTO ESTABLISH MUSIC DAW SOFTWARE

PRODUCT VISION



“Good idea, make it and sell it to Soundcloud”
- anon datapoint no. 3

Anyone got phone number?

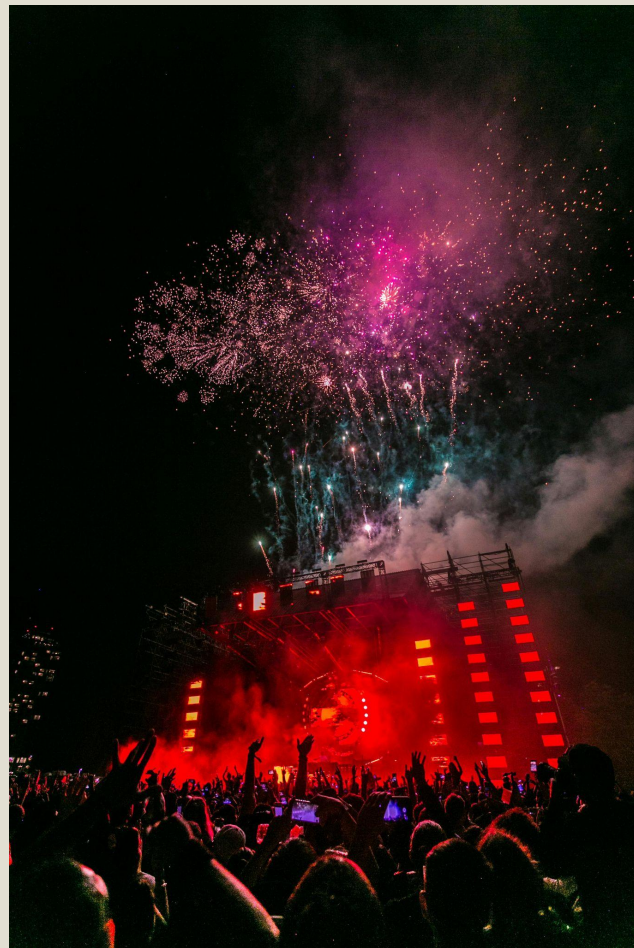
THANK YOU

Special thanks to
Krzysztof Buzar (DSR Mentor)

Dennis Dang for the CRNN base model

and

**Joe, Pino, Dirk, Michael and Ali
for trusting me with their music and
opinions**

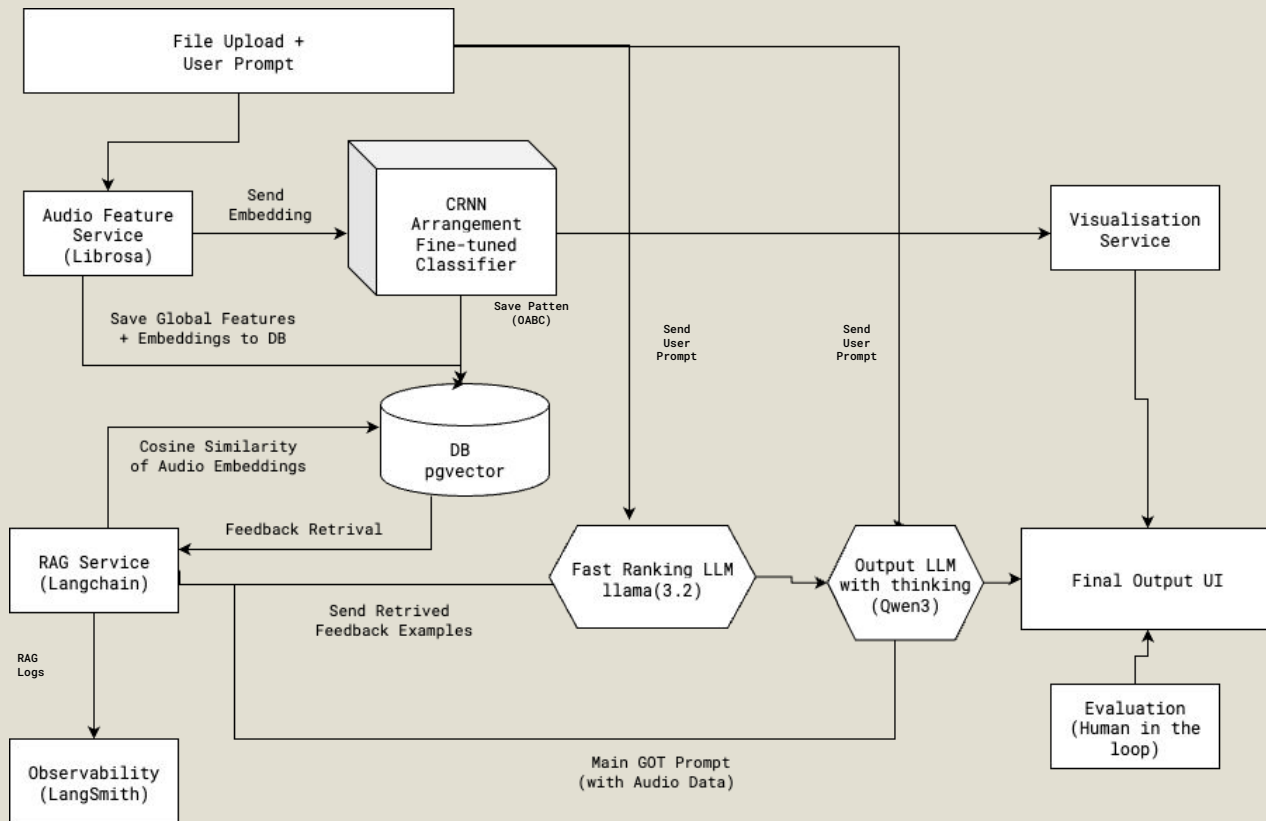


ANY QUESTIONS?

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→ github.com/granttnarg

RESOURCES:

AI MUSIC MENTOR:

<https://github.com/granttnarg/AI-Music-Mentor>

AI TRACKMATE: <https://arxiv.org/abs/2412.06617>

CHORUS DETECTION CRNN:

<https://github.com/dennisvdang/chorus-detection>